

# Engineering Physics (2025)

## Course code 25PY101

### Unit 2: Quantum theories of solids

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# Unit 2 Plan

- 1 Quantum Free Electron Theory
- 2 Fermi-Dirac distribution
- 3 Electronic specific heat of solids
- 4 Density of states (qualitative)
- 5 Success and Failures of quantum free electron theory of solids
- 6 E-k diagram
- 7 Classification of materials based on bands in solids
- 8 Fermi level in semiconductors- intrinsic and extrinsic

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# Quantum states in a cubic box

Consider a particle of mass  $m$  confined in a cubic box of side  $L$  with infinite walls.

- Wavefunction boundary conditions lead to quantum numbers  $n_x, n_y, n_z \in \mathbb{Z}^+$ .

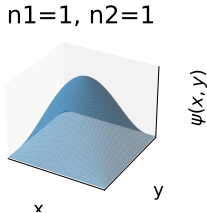
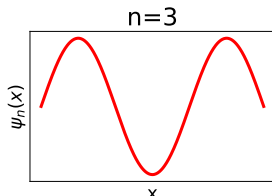
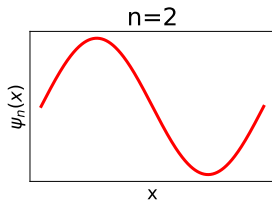
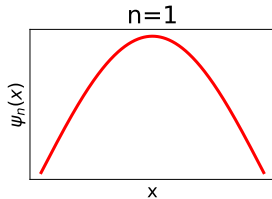
$$\psi_{n_x, n_y, n_z}(x, y, z) = \sqrt{\frac{8}{a^3}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right) \sin\left(\frac{n_z \pi z}{a}\right), \quad n_x, n_y, n_z \in \mathbb{Z}^+$$

- Energy eigenvalues (exact):

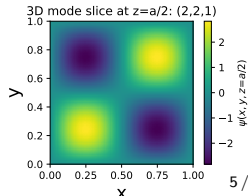
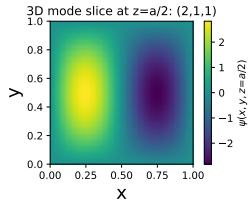
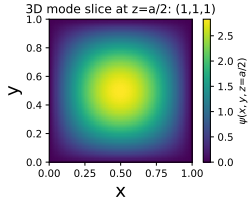
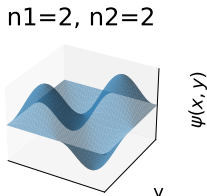
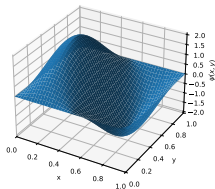
$$E_{n_x, n_y, n_z} = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2).$$

- Define  $n \equiv \sqrt{n_x^2 + n_y^2 + n_z^2}$  (radial coordinate in  $n$ -space).

# Nature of Wavefunctions – Macroscopic atom



2D infinite well (surface) :  $n_1=2, n_2=1$



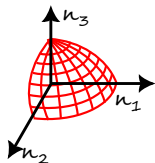
# Counting the number of states up to energy $E$

A state corresponds to a point with positive integer coordinates  $(n_x, n_y, n_z)$  in the positive octant.

- The number of states with  $n$  less than some value  $n_0$  equals the number of integer lattice points in the positive octant inside a sphere of radius  $n_0$ .
- In the continuum (large  $n$ ) approximation we count the volume in  $n$ -space:

$$N(n \leq n_0) \approx g \times \frac{1}{8} \times \frac{4}{3} \pi n_0^3 = g \frac{\pi}{6} n_0^3, =$$

where  $g$  is the degeneracy due to internal degrees of freedom (for electrons,  $g = 2$  for spin).



## Relate $n$ to energy

Using  $E = (h^2/8ma^2)n^2$ , solve for  $n$ :

$$n(E) = \frac{\sqrt{8ma^2E}}{h}.$$

Thus the total number of states with energy less than or equal to  $E$  is

$$N(E) \approx g \frac{\pi}{6} n(E)^3 = g \frac{\pi}{6} \left( \frac{\sqrt{8mL^2E}}{h} \right)^3 = g \frac{\pi}{6} \left( \frac{\sqrt{8mE}}{h} \right)^3 L^3.$$

Here, the volume  $V = L^3$  and  $g=2$ .

# Density of states $Z(E) = \frac{d(\frac{N}{V})}{dE}$

## Definition

- Density of states  $Z(E)$ <sup>a</sup> is defined as the rate of change of the number of states per unit volume upto energy  $E$  with respect to energy  $E$ .
- Therefore, the the number of states per unit volume from energy  $E$  to  $E + dE$  is given by

$$Z(E) dE.$$

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<sup>a</sup>In Neamen, density of states is denoted by  $g(E)$ .

Number of states per unit volume is

$$\rho_N(E) = \frac{N}{V} = \frac{\pi}{3} \left( \frac{\sqrt{8m}}{h} \right)^3 E^{3/2}$$



## DOS in textbook form

Differentiate  $\rho_N(E)$  with respect to  $E$  to get the density of states function:

$$Z(E) = \frac{d\rho_N(E)}{dE} = \frac{\pi}{2} \cdot \left( \frac{\sqrt{8m}}{h} \right)^3 E^{1/2}$$

$$\begin{aligned} Z(E) &= \frac{\pi}{2} \cdot 8 \cdot \left( \frac{2m}{h^2} \right)^{3/2} E^{1/2} \\ &= 4\pi \cdot \left( \frac{2m}{h^2} \right)^{3/2} E^{1/2} \end{aligned}$$

### Key Insight

For 3D infinite potential,  $Z(E) \propto \sqrt{E}$ .

### Problem

*Find the form of  $Z(E)$  for 2D infinite potential.*

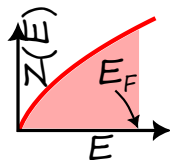
# Fermi level

## Definition

Fermi level of a metal is the maximum energy that an electron can have at  $T = 0$  K.

At absolute zero all states are filled up to the Fermi energy  $E_F$ . The total electron density  $n_c$  (electrons per unit volume) is

$$\begin{aligned} n_c &= \frac{N}{V} = \int_0^{E_F} Z(E) dE = 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \int_0^{E_F} E^{1/2} dE \\ &= 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \cdot \frac{2}{3} E_F^{3/2} \\ &= \frac{8\pi}{3} \left(\frac{2m}{h^2}\right)^{3/2} E_F^{3/2}. \end{aligned}$$



# Relating Fermi energy to electron density

Invert the relation to get the Fermi energy as a function of density:

$$E_F^{3/2} = \left( \frac{h^2}{2m} \right)^{3/2} \cdot \frac{3n_c}{8\pi}.$$

$$\therefore E_F = \frac{h^2}{2m} \left( \frac{3n_c}{8\pi} \right)^{2/3}.$$

This is the Fermi energy of a free electron gas in three dimensions at  $T = 0$ .

## Problem

*Calculate  $E_F$  for metal with electron density  $n_c = 5.8 \times 10^{28} \text{ m}^{-3}$ .*

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# Merits and demerits of QFET

## Merits

- Electrical conductivity
- Thermal conductivity
- Heat capacity

## Demerits

- Cannot explain classification of condensed matter into metals, semiconductors and insulators
- Occurrence of positive Hall coefficient in some metals like Zn.



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## $E - k$ diagram – 1D case

- In a 1D metal, the electron is free as well as independent.
- For a free electron, the energy is related to the momentum  $p$  by

$$E = \frac{p^2}{2m}.$$

- However, since the electron behaves also as a wave, its wavelength  $\lambda$  is related to the momentum by the de Broglie relation

$$\lambda = \frac{h}{p} \quad \Rightarrow \quad p = \frac{h}{\lambda}.$$

- By definition, the wavevector  $k$  of the electron is inversely related to the wavelength as

$$k = \frac{2\pi}{\lambda}$$

- Therefore, the energy is related to the wavevector by

$$E = \frac{1}{2m} \left( \frac{h}{\lambda} \right)^2 = \frac{1}{2m} \left( \frac{hk}{2\pi} \right)^2 = \frac{\hbar^2 k^2}{2m}.$$

## $E - k$ diagram – 3D case

- The analysis for 1D case can be extended to the 3D case by considering the components of momentum  $p_x, p_y, p_z$

$$\vec{p} = p_x \hat{i} + p_y \hat{j} + p_z \hat{k}$$

so that

$$E = \frac{p_x^2 + p_y^2 + p_z^2}{2m}.$$

- The wavevector has three components  $k_x, k_y, k_z$  so that

$$\vec{k} = k_x \hat{i} + k_y \hat{j} + k_z \hat{k}.$$

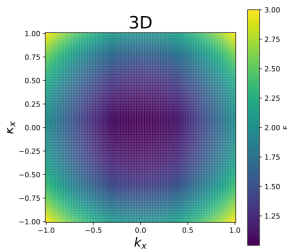
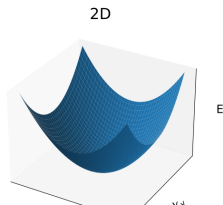
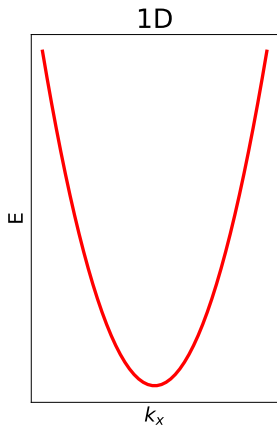
- The energy of the free electron is given by

$$E = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2)$$

### Problem

Fermi wavevector  $k_F$  If the Fermi energy is  $E_F = 5 \text{ eV}$ , find the Fermi wavevector

# Form of $E - k$ diagram – 1D case



## Key Insight

Form of  $E - k$  is parabolic.



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